Birmingham River Champion's
Urban Riverfly Guidance

Urban Riverfly Monitoring is a citizen science initiative where trained volunteers monitor underwater macroinvertebrates. This is one of the four techniques River Champions use to monitor the health of rivers.

The invertebrates (spineless animals) volunteers are looking for are great 'bioindicators', meaning that the presence or absence of different species can indicate different issues (e.g., the absence of a sensitive species may be a sign of pollution). Monitoring these species can provide early warning signs for more serious problems.

Urban Riverfly should take up to 1 hour and sampling should, ideally, be performed every month when volunteers are available and conditions are safe for monitoring to take place. The sampling should be repeated at the same site and, if possible, by the same surveyor. Don't forget to let the River Champions Team know the location of your surveying.

For biosecurity purposes, groups sampling multiple locations (and particularly multiple rivers) on the same day should change surveyors. This prevents invasive species and pollutants from being spread across multiple locations.

Site Identification



Riffle habitats in rivers, characterised by shallow depths, fast flow velocities, coarse sediments (like gravels and cobbles) and 'wave-like' disturbances on the water surface

Before undertaking Urban Riverfly, you should identify a section of river that is easy to access and shallow (below welly height).

Ideally, you should sample in habitats called 'riffles' - shallow waters containing fast flows and larger sediments with 'wave' like features on the water surface.

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Equipment

You will be provided with a sampling kit. This kit will include a D Frame net, bucket, sample tray, sorting tray, spoon, pipet and paintbrush, and magnifying glass, all pictured below.



Kick sampling and hand search

To carry out the kick sample, take your D-fram net and enter the river at your chosen location. Kick sampling involves disturbing riverbed habitats (e.g., plants, algae, sediments) using the base of your foot in a 'kicking' motion. Stand side on to the flow of the river with your net positioned upstream (i.e. the river flowing from you into your net), then kick the riverbed to cause disturbance (you do not need to 'score goals' and kick into the net as this will clog up your tray). Do this for a total of 3 minutes across different locations.

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Some key things to remember:

- You should spend approximately 15-seconds kicking in each spot.
- You should only time when actively kicking.
- You should divide your time up proportionally based on the habitats present (e.g., if your river has equal gravels and plant coverage, you'd spend 1.5-minutes kicking in each).

After the kick sample, you should perform a 1-minute hand search. For this, you should inspect large objects (e.g., wood or rocks) that cannot be disturbed by kicking, and the river margins for 'surface-dwelling' macroinverbrates (e.g., "pond skaters") and add these to your net (it is okay if you cannot see anything during this period).

Macroinvertebrate Sampling

You should either empty your net into a straight into a tray containing river water, or into the bucket containing river water if you have a large amount of material.



If you use a bucket, empty the sample into your tray in small amounts. You should ideally be able to see at least 30% of the white tray (you can wash off and remove large items likes rocks and leaves as necessary). See the image to the left for what this should look like in the tray.

You should record the number of each identified specimen into one of the following abundance categories: **1-9**; **10-99**; **100-999**; **≥1000**. If splitting one sample across multiple trays, you should roughly tally the number of specimens in each tray before coming up with a final abundance category value based on all the trays combined. Urban Riverfly records can be uploaded via our online form, where we will generate summary statistics based that will be displayed on our website.

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If you have one, use the Urban Riverfly monitoring booklet for support with identification. The following resources may be helpful tools for identification!

'Cheats' for Identifying Key Species

Species	Identification Information
Cased caddisflies	Look out for small stones arranged in a 'tube' structure, but only record if you can see the caddisfly inside. Other cases can be made out of plants and even snail shells.
Caseless caddisflies	No tails, but posterior hooks. Look out for a hard head and soft body. Two main types: • Green sedge' caddisflies - green, spikey gills on the side of body; • 'Net-spinning' caddisflies - dark brown head / light brown body, 'broom' like hooks, gills under body
Stoneflies	Both stoneflies and mayflies have 6 legs, but theor tails are different; Stoneflies = 2 tails; Mayflies = 3 tails – but mayfly tails are lost easily, not to be mistaken for stoneflies. Also don't confuse with damselflies (see decision tree below). These 3 species are uncommon in Birmingham's rivers (pollution sensitive), so do not record unless you have sampled at the site before or have confirmation from the BRC project team.
Green drake (burrowing) mayfly	
Flat-bodied (stone clinger) mayfly	
Olive mayfly	The most common mayfly. Moves rapidly (like a torpedo), slimmer body, 'leaflike' gills on side. Don't confuse with blue-winged olives!`
Blue-winged olive mayfly	Darker, wider body than olive mayfly, and have stripes. Tail can stick up like a scorpion. Moves slowly, gills not sticking out.
Freshwater shrimp	Lots of legs, swims on side. Generally peach or cream colour. Contact BRC project team if you see 'striped' pattern or spikes near the appendage as this may be the invasive 'demon' shrimp.
Freshwater hoglouse	Lots of legs, crawls on front. Dark brown.
Adult beetles	Most common will be jet black all over and small. If confident, see the decision tree below on beetle larvae.
Freshwater worms	Generally pink colour. Unlike non-biting midges, worms have no visible head and don't swim, also much longer / wriggly.
Freshwater snails	Do not record empty shells. May come in different forms besides more 'typical' looking snails, such as 'ramshorn' snails (coiled and flat) and 'limpets' (wizards hat).
Freshwater leeches	2 key types. One much larger, darker that moves by expanding/contracting (elastic band). The other smaller, curl up in cshape and have 'train track' pattern on back.

River Fly Identification

Legs?
NO (or not visible for caseless caddisflies)

External shell/case?

Yes

Brittle case?

Yes

Snails (Gastropoda)



No (case of stone or plants)

Cased cassidfly (Trichoptera)

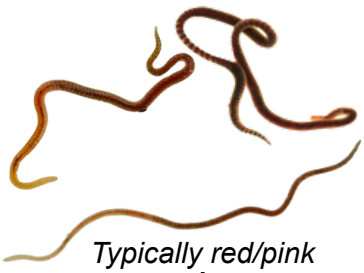


No

Lack of structure (long and wiggly)?

Yes

Worm (Oligochaeta)



Typically red/pink

No

Distinct head?

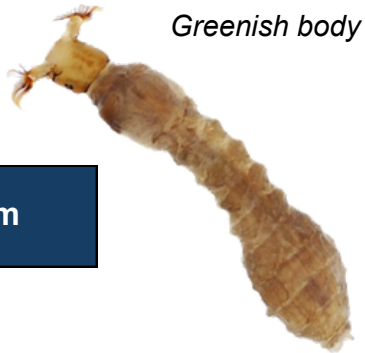
Yes

Swollen abdomen?

Yes

Blackfly larvae (Simuliidae)

Greenish body



No

Non-biting midge larvae

Synchronised side swimming



Don't mix up

No

Suction pad?

Yes

Leech (Hirudinae)



Elastic band movement



train track pattern

No

Flatworm



often have 'horn-like' features

Legs? **YES**

Swins on and not flattened top to bottom?

>6

No. of legs?

6

No. of tails?

Yes

No

Striped pattern and spikes on back?

Hoglouse (Asellidae)



No

Yes

Native shrimp (Gammaridae)

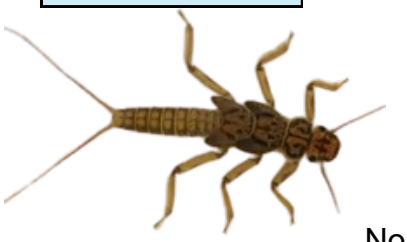


Invasive "demon" or "killer" Shrimp



2

Stonefly (Cloeoptera)



0

Posterior hooks?

3

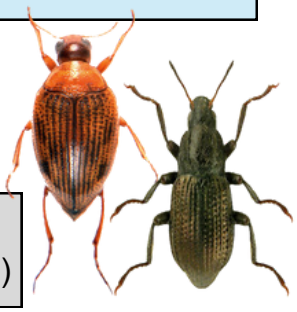
No

Yes

Beetle (Coleoptera)

Beetle larvae (record if confident)

No gills and 'hard' body same colour as head





Mayflies (Ephemeroptera)

Don't Mix Up

Don't mix up

Pulsing gills covering back?

Caseless caddisflies (Trichoptera)



Danselfly larvae

No

Yes

Squat arms?



Plate like tail, not hair like

No

Yes

Long torpedo movement?



Hard head soft body

Yes

No

Green drake mayfly (Ephemeridae)

Olive mayfly

Blue winged olive mayfly

Flat bodied stone clinger (Heptageniidae)



Pointed head 'burrowing mayfly'

Slender body and torpedo movement. Leaf gills on side of body





More erratic swimming, striped bodypattern, gills over body

'Freshwater Ferrari'

